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1 ASTAR Middleware Full Deployment Guide: Mac to Windows 11 Server

This guide covers the full field workflow:

1. Connect from a Mac to a Windows 11 server.
2. Copy the ASTAR middleware ZIP package to the server.
3. Install the middleware on the server.
4. Prepare configuration, network access, and tool prerequisites.
5. Validate, test, deploy, and run the middleware as a Windows service.

The target server is assumed to be Windows 11. Commands in Windows sections are PowerShell commands.

1.1 1. What Will Be Installed

Item	Value
Windows install root	C:\SECSGEM_EAP
Application folder	C:\SECSGEM_EAP\app
Main config	C:\SECSGEM_EAP\app\config\production.yaml
Logs	C:\SECSGEM_EAP\logs
Data/outbox	C:\SECSGEM_EAP\data
Service name	AstarSecsGemEapMiddleware
Local CSV output	D:\MachineData\EAP_<display_name>\csv_in
Default HTTPS upstream	https://astar-monitoring.linkstuffs.com

The middleware connects to tools over HSMS/SECS/GEM, writes per-lot CSV files, and sends telemetry to Linkstuffs / ThingsBoard.

1.2 2. Information Needed Before Deployment

Collect this before opening the server.

1.2.1 Windows Server Access

- Windows server IP address or hostname.
- Windows username and password.
- Confirmation that the account has local Administrator access.
- Confirmation that Remote Desktop is enabled on the Windows server.

1.2.2 Per Tool

Field	Example	Notes
endpoint_id	T00L_02	Internal middleware ID.
display_name	DAVINCI200_MC4_HC1_01	Must match Linkstuffs token key.
machine_profile	davinci_200_mc4_hc1	Supported: davinci_200_mc4_hc1, spts_fxp_omega,ptiq_secsgem.
Tool HSMS IP	10.10.20.32	Tool IP reachable from Windows server.
Tool HSMS port	5000	Confirm with tool owner/vendor.
secs_device_id	0	Must exactly match the tool's HSMS Device ID.
HSMS mode	active	Normal case: tool is passive, middleware connects to it.
Linkstuffs device token	...	One token per device for HTTPS mode.

Important: only one host can normally hold an HSMS connection to a tool. If MES or vendor software is already connected, disconnect it before testing this middleware.

1.3 3. Prepare the Deployment ZIP on the Mac

From the repository root on the Mac:

```
cd /Users/nrzngr/Desktop/astar-middleware-main
./scripts/build_deploy_package.sh
```

The script creates:

deploy_out/astar-middleware-deploy-YYYY-MM-DD.zip

Use the newest ZIP in deploy_out.

The ZIP contains:

- install.ps1
 - README_DEPLOY.txt
 - SETUP_CHECKLIST.txt
 - QUICKSTART.md
 - source\
 - wheels\
 - python\ if the offline Python installer is available
-

1.4 4. Connect to the Windows 11 Server from Mac

1.4.1 Option A: Microsoft Remote Desktop

1. Install **Microsoft Windows App** or **Microsoft Remote Desktop** from the Mac App Store.
2. Open the app.
3. Add a PC:
 - PC name: Windows server IP or hostname.
 - User account: Windows username and password.
4. Connect.
5. Accept the certificate warning if this is the expected internal server.

1.4.2 Option B: Finder SMB Share

Use this if file sharing is enabled on the Windows server.

1. In Finder, press Cmd+K.
2. Enter:

```
smb://<windows-server-ip>/C$
```

or a shared folder path:

```
smb://<windows-server-ip>/<share-name>
```

3. Log in with a Windows Administrator account.
4. Copy the deployment ZIP to a simple path such as:

```
C:\Users\<windows-user>\Downloads
```

1.4.3 Option C: RDP Shared Folder

If using Microsoft Remote Desktop:

1. Edit the saved PC connection.
2. Open **Folders**.
3. Add a Mac folder, for example:

```
/Users/nrzngr/Desktop/astar-middleware-main/deploy_out
```

4. Reconnect to the Windows server.
5. Inside Windows, open File Explorer.
6. The shared Mac folder appears under redirected drives or devices.
7. Copy the ZIP to:

C:\Users\<windows-user>\Downloads

1.4.4 Option D: SCP

Use this only if OpenSSH Server is enabled on Windows.

From the Mac:

```
scp deploy_out/astar-middleware-deploy-YYYY-MM-DD.zip <windows-user>@<windows-server-ip>:/C:/Users
```

1.5 5. Extract the ZIP on Windows

On the Windows server:

1. Open File Explorer.
2. Go to:

C:\Users\<windows-user>\Downloads

3. Right-click the ZIP.
4. Choose **Extract All**.
5. Extract to:

C:\Users\<windows-user>\Downloads\astar-middleware-deploy

After extraction, confirm this file exists:

C:\Users\<windows-user>\Downloads\astar-middleware-deploy\install.ps1

1.6 6. Install the Middleware

Open **PowerShell as Administrator**.

Allow local scripts once:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope LocalMachine
```

Go to the extracted package:

```
cd C:\Users\<windows-user>\Downloads\astar-middleware-deploy
```

Run the installer:

```
.\install.ps1
```

Expected installer behavior:

- Checks or installs the required Python version.
- Creates C:\SECSGEM_EAP.
- Copies source files to C:\SECSGEM_EAP\app.
- Installs Python dependencies from the bundled wheels folder.
- Runs an import smoke test.
- Adds outbound Windows Firewall rules for HSMS and Linkstuffs MQTT.
- Opens production.yaml in Notepad.

Expected successful output includes:

Installation complete.

If Windows SmartScreen blocks the script:

1. Click **More info**.
2. Click **Run anyway**.

If PowerShell says the script is blocked, right-click `install.ps1`, choose **Properties**, tick **Unblock**, then run the script again.

1.7 7. Configure the Middleware

Open the config:

```
notepad C:\SECSGEM_EAP\app\config\production.yaml
```

1.7.1 Configure Linkstuffs HTTPS

For Cloudflare or normal HTTPS deployments, keep MQTT disabled and HTTPS enabled:

```
linkstuffs:
  enabled: false

linkstuffs_http:
  enabled: true
  base_url: "https://astar-monitoring.linkstuffs.com"
  device_tokens:
    SPTS_fxP_OMEGA_01: ""
    DAVINCI200_MC4_HC1_01: "PASTE-DAVINCI-TOKEN-HERE"
    PTIQ_01: ""
  verify_tls: true
```

Rules:

- The key under `device_tokens` must exactly match the machine `display_name`.
- Empty token means that device is skipped for HTTPS publishing.
- A 401 Unauthorized during testing usually means the token is wrong.

1.7.2 Configure Machines

Example DaVinci machine:

```
machines:
  - endpoint_id: "T00L_02"
    display_name: "DAVINCI200_MC4_HC1_01"
    machine_profile: "davinci_200_mc4_hc1"
    host: "10.10.20.32"
    port: 5000
    secs_device_id: 0
    hsms_mode: "active"
    enabled: true
    enable_alarms: false
    request_online: false
    drain_spool_on_connect: false
    local_csv_path: "D:/MachineData/EAP_DAVINCI200_MC4_HC1_01/csv_in"
    network_csv_path: "\\TD-DATASVR-F2C4\\TD_DAVINCI200_MC4_HC1_01.csv_in"
    admin_config_path: "C:/SECSGEM_EAP/machines/DAVINCI200_MC4_HC1_01/config"
```

Set unused tools to:

```
enabled: false
```

Save the file.

1.8 8. Deployment Preparation Checks

Run these before the first live deployment.

1.8.1 Check Python

```
python --version
```

The version must match the package's `PYTHON_VERSION.txt`, normally Python 3.11 for the bundled offline package.

1.8.2 Check Tool Network Reachability

```
ping 10.10.20.32
```

If ping is disabled by policy, use PowerShell TCP testing:

```
Test-NetConnection 10.10.20.32 -Port 5000
```

1.8.3 Check D:\ CSV Drive

```
Test-Path D:\
```

Expected:

True

If D:\ does not exist, either create/provision the drive or change `local_csv_path` in `production.yaml`.

1.8.4 Check Network CSV Share

If `network_csv_path` is enabled:

```
Test-Path "\\TD-DATASVR-F2C4\TD_DAVINCI200_MC4_HC1_01.csv_in"
```

If the share is unavailable, the middleware still keeps the local CSV, but the network mirror will fail until the share is fixed.

1.8.5 Confirm Tool-Side Settings

With the tool owner/vendor, confirm:

- Tool HSMS is passive if middleware `hsms_mode` is active.
 - Tool IP, port, and SECS Device ID are correct.
 - No other host is connected to the tool's HSMS port.
 - For DaVinci/FabLink, event reporting style is acceptable. E30/S6F11 gives full collection-event detail. E40 notifications are ingested but are coarser.
 - `enable_alarms: true` is used only if the tool owner confirms the tool accepts host-side S5F3 alarm enable.
-

1.9 9. Validate the Configuration

Open a new PowerShell window:

```
cd C:\SECSGEM_EAP\app
python -m eap_middleware validate-config --config config\production.yaml
```

Expected result includes:

"valid": true

Fix all validation errors before continuing.

1.10 10. Test Machine Connectivity

Test one machine:

```
cd C:\SECSGEM_EAP\app
python -m eap_middleware test-machine --config config\production.yaml --endpoint-id TOOL_02
```

Test all enabled machines:

```
python -m eap_middleware test-machine --config config\production.yaml --endpoint-id ALL
```

Common results:

Error	Meaning	Fix
Connection refused	Wrong IP/port, tool HSMS is off, or another host is connected.	Confirm tool HSMS service and disconnect other host.
Connection timeout	Network path blocked.	Check VLAN, firewall, route, and port.
Device ID mismatch	secs_device_id does not match tool.	Correct secs_device_id in production.yaml.

1.11 11. Test Linkstuffs / ThingsBoard

```
cd C:\SECSGEM_EAP\app
python -m eap_middleware test-linkstuffs --config config\production.yaml
```

If this fails:

- Confirm linkstuffs_http.enabled: true.
 - Confirm base_url.
 - Confirm the token for the exact display_name.
 - Confirm outbound HTTPS port 443 is allowed.
-

1.12 12. First Live Run Before Service Deployment

Run the middleware interactively first:

```
cd C:\SECSGEM_EAP\app
python -m eap_middleware run-service --config config\production.yaml
```

Watch for:

- Tool connection established.
- No repeated reconnect loop.
- Linkstuffs publish success.
- CSV files created under D:\MachineData\EAP_<display_name>\csv_in.

Trigger or wait for a real lot event, then confirm:

```
Get-ChildItem D:\MachineData\EAP_DAVINCI200_MC4_HC1_01\csv_in
```

Stop the interactive run with:

Ctrl+C

1.13 13. Install as a Windows Service

NSSM is recommended for production because it auto-starts on boot and captures logs cleanly.

1.13.1 Install NSSM

Copy `nssm.exe` to:

`C:\Tools\nssm\nssm.exe`

Make sure `nssm.exe` is on PATH or pass its full path to the service installer.

1.13.2 Register the Service

Open **PowerShell as Administrator**:

```
cd C:\SECSGEM_EAP\app
scripts\install_service.ps1 -NssmExe "C:\Tools\nssm\nssm.exe"
```

If Python is not auto-detected:

```
scripts\install_service.ps1 `
  -PythonExe "C:\Python311\python.exe" `
  -NssmExe "C:\Tools\nssm\nssm.exe"
```

Start the service:

```
Start-Service AstarSecsGemEapMiddleware
```

Check status:

```
Get-Service AstarSecsGemEapMiddleware
```

Expected:

Status	Name
Running	AstarSecsGemEapMiddleware

1.14 14. Verify the Full Deployment

Watch logs:

```
Get-Content C:\SECSGEM_EAP\logs\service_stderr.log -Tail 50 -Wait
```

Check service stdout if needed:

```
Get-Content C:\SECSGEM_EAP\logs\service_stdout.log -Tail 50 -Wait
```

Confirm:

- Service is running.
 - Each enabled tool connects.
 - No repeated validation/config errors.
 - Telemetry appears in Linkstuffs.
 - CSV appears in `D:\MachineData\EAP_<display_name>\csv_in`.
 - If configured, CSV also appears in the network share path.
-

1.15 15. Normal Operations Commands

`Start-Service` AstarSecsGemEapMiddleware
`Stop-Service` AstarSecsGemEapMiddleware
`Restart-Service` AstarSecsGemEapMiddleware
`Get-Service` AstarSecsGemEapMiddleware

Validate after config edits:

```
cd C:\SECSGEM_EAP\app
python -m eap_middleware validate-config --config config\production.yaml
```

Test all machines:

```
python -m eap_middleware test-machine --config config\production.yaml --endpoint-id ALL
```

Test upstream:

```
python -m eap_middleware test-linkstuffs --config config\production.yaml
```

1.16 16. Updating an Existing Deployment

1. Build a fresh ZIP on the Mac:

```
cd /Users/nrzngr/Desktop/astar-middleware-main
./scripts/build_deploy_package.sh
```

2. Copy the new ZIP to the Windows server.
3. Extract it.
4. Stop the service:

`Stop-Service` AstarSecsGemEapMiddleware

5. Run installer as Administrator:

```
cd C:\Users\<windows-user>\Downloads\astar-middleware-deploy
.\install.ps1
```

The installer preserves the existing live config:

```
C:\SECSGEM_EAP\app\config\production.yaml
```

It writes the new template as:

```
C:\SECSGEM_EAP\app\config\production.yaml.new
```

and saves a backup snapshot as:

```
C:\SECSGEM_EAP\app\config\production.yaml.bak
```

6. Compare and merge any needed new config fields:

```
notepad C:\SECSGEM_EAP\app\config\production.yaml
notepad C:\SECSGEM_EAP\app\config\production.yaml.new
```

7. Validate:

```
cd C:\SECSGEM_EAP\app
python -m eap_middleware validate-config --config config\production.yaml
```

8. Start the service:

`Start-Service` AstarSecsGemEapMiddleware

9. Watch logs:

```
Get-Content C:\SECSGEM_EAP\logs\service_stderr.log -Tail 50 -Wait
```

1.17 17. Rollback

If the previous deployment worked and the new deployment fails:

1. Stop the service:

`Stop-Service AstarSecsGemEapMiddleware`

2. Restore the previous config if needed:

`Copy-Item C:\SECSGEM_EAP\app\config\production.yaml.bak C:\SECSGEM_EAP\app\config\production.yaml`

3. Reinstall from the previous known-good ZIP if code rollback is required.
4. Validate:

`cd C:\SECSGEM_EAP\app`

`python -m eap_middleware validate-config --config config\production.yaml`

5. Start service:

`Start-Service AstarSecsGemEapMiddleware`

1.18 18. Troubleshooting

1.18.1 Installer Cannot Run

Run PowerShell as Administrator:

`Set-ExecutionPolicy RemoteSigned -Scope LocalMachine`

If the file is blocked:

`Unblock-File .\install.ps1`

1.18.2 Python Version Error

Check:

`python --version`

`type .\PYTHON_VERSION.txt`

The installed Python version must match the package's required version.

1.18.3 Machine Test Fails

Check:

`Test-NetConnection <tool-ip> -Port <tool-port>`

Then confirm:

- Tool HSMS service is running.
- No other host is connected.
- secs_device_id matches the tool.
- Firewall/VLAN routing allows the connection.

1.18.4 Linkstuffs Test Fails

Check:

- linkstuffs_http.enabled is true.
- base_url is correct.
- Token matches the exact display_name.
- Server can access HTTPS port 443.

1.18.5 Service Starts Then Stops

Read logs:

```
Get-Content C:\SECSGEM_EAP\logs\service_stderr.log -Tail 100
Get-Content C:\SECSGEM_EAP\logs\service_stdout.log -Tail 100
```

Also check Windows Event Viewer:

Start menu -> Event Viewer -> Windows Logs -> Application

1.18.6 No CSV Output

Confirm:

- A real lot/tool event occurred.
- The machine is connected.
- D:\ exists.
- local_csv_path is correct.
- The Windows service account has write permission.

1.18.7 No Telemetry in Linkstuffs

Confirm:

- test-linkstuffs passes.
 - The token is for the correct device.
 - The device name in Linkstuffs matches display_name.
 - The logs show publish attempts and no HTTP errors.
-

1.19 19. Final Deployment Checklist

- ☐ ZIP built on Mac.
- ☐ Remote Desktop access works from Mac to Windows server.
- ☐ ZIP copied to Windows server.
- ☐ ZIP extracted.
- ☐ install.ps1 completed successfully.
- ☐ production.yaml edited.
- ☐ validate-config returns valid.
- ☐ test-machine passes for each enabled tool.
- ☐ test-linkstuffs passes.
- ☐ Interactive run-service works.
- ☐ Windows service installed with NSSM.
- ☐ Service is running.
- ☐ Logs show successful tool connection.
- ☐ CSV output confirmed.
- ☐ Linkstuffs telemetry confirmed.